10. $\int_{1}^{e} x^3 \ln x \, dx$ 의 값은? [3점]

① $\frac{3e^4}{16}$ ② $\frac{3e^4+1}{16}$ ③ $\frac{3e^4+2}{16}$

④ $\frac{3e^4+3}{16}$ ⑤ $\frac{3e^4+4}{16}$